

- Q.22 Describe moving coil instruments?
 Q.23 How the Aircrafts engines are Started?
 Q.24 What is a CHT circuit? Where are they used?
 Q.25 Explain the working of a current limiter.
 Q.26 Write in brief about Aircraft Wiring system.
 Q.27 What is the role of a slip ring?
 Q.28 Write in brief the functioning of inter poles.
 Q.29 How does a revolving field type AC generator work?
 Q.30 What are the applications of Single Phase and three Phase AC motor.
 Q.31 How is paralleling of generators done?
 Q.32 What is the potential xmer?
 Q.33 Describe landing light circuit.
 Q.34 What is the procedure for maintenance of installed batteries?
 Q.35 What do you mean by a signal filter? Give one example

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Explain the working of a DC motor and AC generator.
 Q.37 Describe special circuits for safety and operations.
 Q.38 Name the various types of measuring instruments. How lead acid and Ni-Cd batteries are charged?

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5th Sem / AME Sub: Aircraft Electrical Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A piezoelectric transducer has a _____.
 a) Very high sensitivity b) Low sensitivity
 c) High sensitivity d) Zero sensitivity
 Q.2 Identify the principle behind the working of an A.C. generator.
 a) Eddy currents
 b) Faraday's law
 c) Lenz's law
 d) Electromagnetic induction
 Q.3 In the cables, the location of fault is usually found out by comparing
 a) The resistance of the conductor
 b) The inductance of conductors
 c) The capacitances of insulated conductors
 d) All above parameters
 Q.4 Which among the following is very fast fuse?
 a) Kitkat b) Semiconductor
 c) Copper brushes d) Field magnet

- Q.5 Thermistors have ____
- Positive temperature coefficient
 - Negative temperature coefficient
 - Zero temperature coefficient
 - Infinite temperature coefficient
- Q.6 Slip rings are used in ____.
- Battery
 - AC Generator
 - DC Generator
 - None of the above
- Q.7 What is the full form of MCB?
- Miniature contact breaker
 - Mini circuit breaker
 - Miniature circuit breaker
 - Mini contact breaker
- Q.8 When a compressive force is applied to a quartz crystal then ____.
- Positive charges are induced
 - Negative charges are induced
 - No charge is induced
 - Both positive and negative charges are induced.
- Q.9 AC bridge is an outcome of ____.
- Kelvin bridge
 - Megger
 - De Sauty bridge
 - Wheatstone bridge

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- Q.10 In a DC generator the ripples in the direct emf generated can be reduced by ____
- Using conductor of annealed copper
 - Using commutator with large number of segments
 - Using carbon brushes of superior quality
 - Using equalizer rings

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What do you mean by an instrument?
- Q.12 What is frequency meter?
- Q.13 Describe lacing.
- Q.14 What is the difference between circuit breaker and current breaker?
- Q.15 What are the protective devices?
- Q.16 What is corona threshold?
- Q.17 What is the purpose of a static discharge wick?
- Q.18 Why the routing of wire bundles needed?
- Q.19 What are types of voltage regulators?
- Q.20 What actually fuel content indicator measures?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the purpose of indicating instruments?

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